

Bode Diagrams

Adding up simple factors on a log-frequency axis.

Ogata, *Modern Control Engineering*, Ch. 7.

In this tutorial you will:

- learn the asymptotic rules for gain, integrator, and pole/zero factors,
- compare an asymptotic sketch with the exact **bode** plot, and
- read the gain and phase crossover frequencies with **margin**.

Step through with **Ctrl+Enter**, or render a report with **publish**.

A Bode diagram plots $20 \log_{10} |G(j\omega)|$ (dB) and $\angle G(j\omega)$ (deg) vs. $\log_{10} \omega$. Because $G(j\omega)$ is normally a product of simple pole/zero/gain/integrator factors, and $\log |G_1 G_2| = \log |G_1| + \log |G_2|$, the log-magnitude and phase of each factor can be sketched separately on a log-frequency axis and simply **added** – this is the key advantage of the logarithmic plot.

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Standard (Bode) Factors and Their Asymptotic Rules

1. **Gain** K : contributes a constant $20 \log_{10} |K|$ dB, 0° (or 180° if $K < 0$) phase, independent of ω .
2. **Integrator/differentiator** $(j\omega)^{\pm 1}$: magnitude slope of ∓ 20 dB/decade through 0 dB at $\omega = 1$; constant phase of $\mp 90^\circ$.
3. **First-order pole/zero** $(1+j\omega T)^{\pm 1}$: low-frequency asymptote 0 dB (flat); high-frequency asymptote slope ± 20 dB/decade beyond the **corner frequency** $\omega = 1/T$; phase goes from 0° to $\pm 90^\circ$, passing through $\pm 45^\circ$ at the corner.
4. **Quadratic pole/zero** $\left(1 + 2\zeta(j\omega/\omega_n) + (j\omega/\omega_n)^2\right)^{\pm 1}$: asymptotic slope ± 40 dB/decade beyond ω_n ; near the corner the actual curve deviates from the asymptote by an amount depending on ζ (resonant peak for small ζ); phase goes from 0° to $\mp 180^\circ$.

Worked Example: Sketch vs. Exact Bode Plot

$G(s) = \frac{10(s+2)}{s(s+10)}$. Rewrite in Bode (time-constant) form:

$$G(j\omega) = \frac{10 \times 2(1+j\omega/2)}{j\omega \times 10(1+j\omega/10)} = \frac{2(1+j\omega/2)}{j\omega(1+j\omega/10)}$$

Corner frequencies: zero at $\omega = 2$, pole at $\omega = 10$, plus an integrator (pole at the origin).

```
K_bode = 2;           % Bode gain after factoring to time-constant form
z1 = 2; p1 = 10;      % corner frequencies

w = logspace(-2,3,500);
s = 1j*w;

% Exact response
```

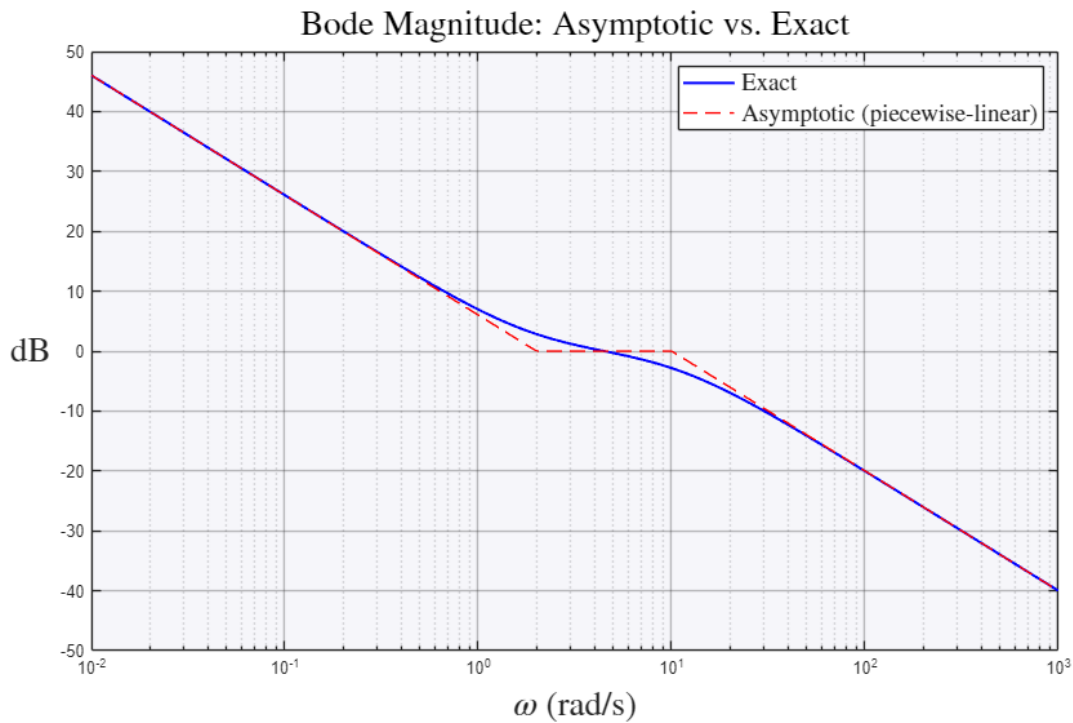
```

G = tf([10 20],[1 10 0]);
[mag,phase] = bode(G,w);
mag = squeeze(mag); phase = squeeze(phase);

% Asymptotic (piecewise-linear) approximation, built term by term:
% gain + integrator (-20 dB/dec), then +20 dB/dec beyond the zero corner
% and -20 dB/dec beyond the pole corner.
mag_dB_pieewise = 20*log10(K_bode) - 20*log10(w);
mag_dB_pieewise = mag_dB_pieewise + 20*log10(max(w/z1,1)) - 20*log10(max(w/p1,1));

figure
semilogx(w,20*log10(mag),'b','LineWidth',1.5)
hold on
semilogx(w,mag_dB_pieewise,'r--')
hold off
legend('Exact','Asymptotic (piecewise-linear)','Interpreter','latex','FontSize',14)
title('Bode Magnitude: Asymptotic vs. Exact','Interpreter','latex','FontSize',20)
ylabel('dB','Interpreter','latex','FontSize',20)
set(get(gca,'YLabel'),'Rotation',0)
xlabel('\omega (rad/s)','Interpreter','latex','FontSize',20)
grid on

```



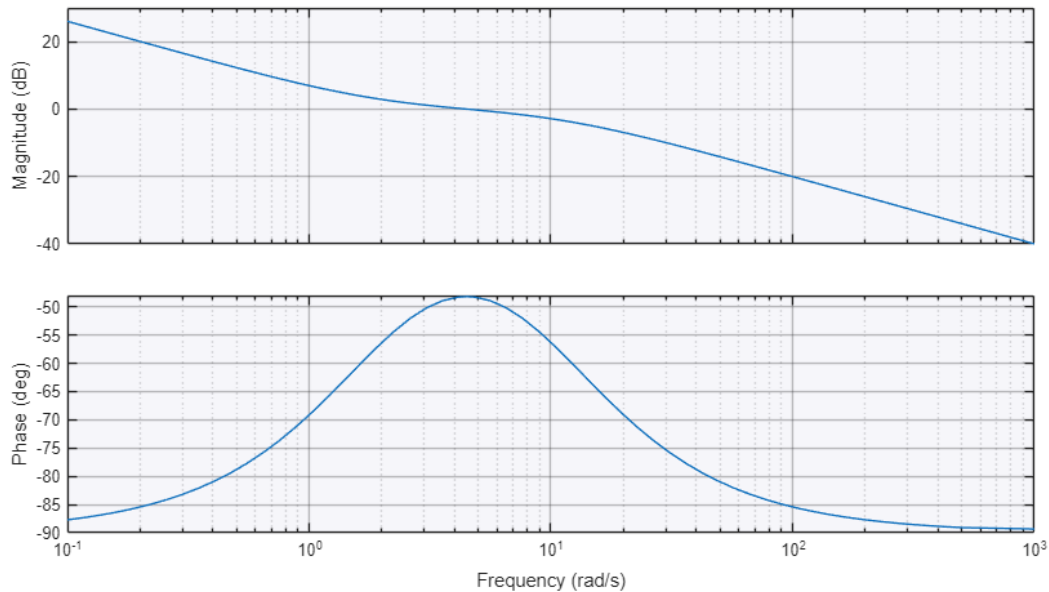
Full Bode Diagram via the Control System Toolbox

```

figure
bode(G)
title('Bode Diagram: $G(s)=\frac{10(s+2)}{s(s+10)}$', 'Interpreter','latex','FontSize',20)
grid on

```

$$\text{Bode Diagram: } G(s) = \frac{10(s+2)}{s(s+10)}$$



Gain Crossover and Phase Crossover Frequencies

The **gain crossover frequency** ω_c is where $|G(j\omega)| = 1$ (0 dB); the **phase crossover frequency** ω_p is where $\angle G(j\omega) = -180^\circ$. These define the gain and phase margins (see `StabilityAnalysis.m`).

```
[Gm,Pm,Wcp,Wcg] = margin(G);
fprintf('Gain crossover frequency wc = %.4f rad/s\n', Wcg)
fprintf('Phase crossover frequency wp = %.4f rad/s\n', Wcp)
```

```
Gain crossover frequency wc = 4.4721 rad/s
Phase crossover frequency wp = NaN rad/s
```

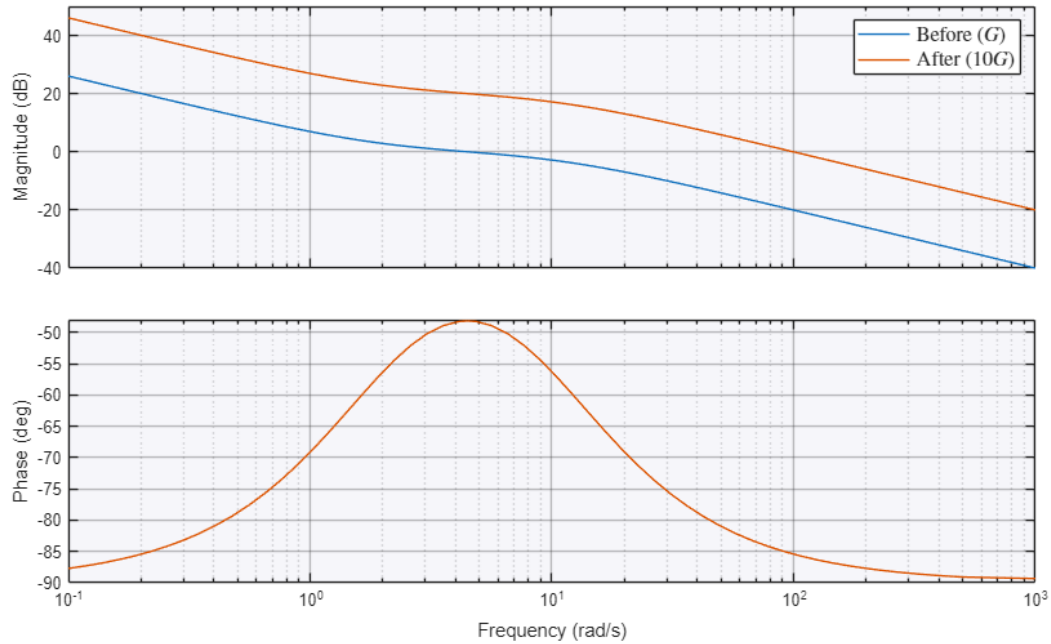
Before vs. After: Raising the Gain Shifts the Magnitude Up

Multiplying the loop gain by 10 lifts the entire magnitude curve by 20 dB, pushing the gain-crossover frequency higher and eroding the phase margin – the frequency-domain ”before/after” of a gain change.

```
figure
bode(G, 10*G)
legend('Before ($G$)', 'After ($10G$)', 'Interpreter', 'latex', 'FontSize', 12)
title('Bode: Before vs. After a 20 dB Gain Increase', 'Interpreter', 'latex', 'FontSize', 16)
grid on
[~,Pm1] = margin(G);
[~,Pm2] = margin(10*G);
fprintf('Phase margin: before = %.1f deg, after (10x gain) = %.1f deg\n', Pm1, Pm2)
```

```
Phase margin: before = 131.8 deg, after (10x gain) = 94.6 deg
```

Bode: Before vs. After a 20 dB Gain Increase



Try it yourself

- Change the multiplier from 10 to 100 and notice the whole magnitude curve lift by 40 dB, dropping the phase margin further.
- Move the zero corner z_1 from 2 to 20 and watch the mid-frequency +20 dB/decade stretch shrink.